

BERNARDO NANDANIAR SUNIA

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NLP Engineer | Data Scientist | Machine Learning Engineer



Informatics graduate (GPA 3.78, Cumlaude) specializing in NLP, data analytics, and applied machine learning, with production experience building chatbots, sentiment analysis pipelines, and multilingual retrieval systems. Skilled in Python, PyTorch, and transformer-based models (BERT, SBERT), with particular depth in Indonesian-language NLP such as sarcasm, complex-negation, code-mixing, and informal text.

EDUCATION

Diponegoro University (Aug 2022 – Mar 2026)

Bachelor of Informatics - GPA 3.78 (Cumlaude) | Graduated March 2026

WORK EXPERIENCE

PT Wiraky Nusa Telekomunikasi – Bandung, Indonesia

Full Time – IT Analyst (May 2026 – Jul 2026)

- Designed and fine-tuned an Aspect Sentiment Quad Extraction (ASQE) model to extract aspects, opinions, categories, and sentiment from 1,000+ self-labeled reviews, with a focus on challenging Indonesian-language cases including sarcasm, negation, code-mixing, and informal text.
- Built and deployed an NLP-based chatbot using embedding similarity matching, with a CRUD admin dashboard and automated data-retention system for efficient knowledge-base management.
- Developed a secure review analytics system combining multi-signal anomaly detection, interactive dashboards, and a property investment analytics pipeline to support data-driven decision-making.

Class II Ambarawa Correctional Facility – Ambarawa, Indonesia

Intern - Automated Information System Chatbot Developer (May 2025 – Jul 2025)

- Led end-to-end chatbot development covering stakeholder interviews, requirements definition, training dataset optimization, QA, and performance monitoring.

Computer Laboratory, Diponegoro University – Semarang, Indonesia

Laboratory Teaching Assistant (Jun 2023 – Jun 2025)

- Mentored 20+ students in Python and algorithmic logic while designing structured assessments to evaluate IT competencies, contributing to a 15% improvement in average lab grades.

East Semarang District – Semarang, Indonesia

Intern - Financial Reporting Application Developer (Dec 2024 – Feb 2025)

- Built a full-stack financial reporting app with ReactJS, JavaScript, and SQL, from requirements gathering to UI development and staff training, improving data accuracy and efficiency operations.

YC Electric – Semarang, Indonesia

Part Time - Assistant Manager, Supply Chain Assistant, and Creative Designer (Jan 2022 – Dec 2024)

- Optimized inventory tracking, streamlined supplier logistics, and produced 30+ promotional materials with digital design tools, boosting brand visibility and marketing impact.

KEY PROJECT

Aspect Sentiment Classification: TF-IDF vs. SBERT on Shopee Reviews (Aug 2025 – Feb 2026)

Technologies: Jupyter Notebook, Scikit-learn, Sentence-Transformers, Pandas, SciPy, Sastrawi, Statsmodels

- Built a labeled dataset from 1,000+ Shopee reviews and compared TF-IDF and SBERT representations paired with Logistic Regression, with SBERT achieving 94% F1-Score.

Custom Search Engine Using VSM and LSI (Mar 2024 – May 2024)

Technologies: Python, Jupyter Notebook, Streamlit, Pandas, NumPy, Sastrawi, NLTK, scikit-learn, Matplotlib

- Developed an interactive custom search engine using VSM and LSI to improve document retrieval, combining text preprocessing, TF-IDF, cosine similarity, and semantic analysis through SVD.

Advanced Information Retrieval System Using ML & Deep Learning (Aug 2024 – Dec 2024)

Technologies: Jupyter Notebook, Streamlit, Pandas, NumPy, mBERT, PyTorch, Matplotlib, Plotly

- Developed a multilingual semantic search system using fine-tuned mBERT and FAISS, enabling efficient retrieval documents through dense embedding-based similarity search.

CERTIFICATES

Intellectual Property Certificate – Software (Ministry of Law, 2026) | Intellectual Property Certificate – Chatbot (Ministry of Law, 2025) | Data Classification and Summarization (IBM SkillsBuild, 2025) | PyTorch & Generative AI (Avalon AI, 2024) | SQL & Java Programming (Oracle Academy, 2024) | Cloud Computing (Alibaba Cloud, 2024) | Data Analysis Bootcamp (Diponegoro University, 2023)

SKILLS

Programming & Tools: Python, Sentence-Transformers, Scikit-learn, FAISS, SQL, Pandas, NumPy, Streamlit, Jupyter, Git, Docker

Methods: NLP, Machine Learning, Information Retrieval, Data Analysis & Visualization, Requirements Gathering, QA Testing, Modeling

Language: English, CEFR 488, Indonesian (Native)